

# FRONTIERS IN COMPUTING: DISCUSSION 1

RENTO SAIJO

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## Cloud infrastructure and service models

- Cloud computing delivers processing power, storage, and applications as services accessible over a network. In the public cloud, providers operate data centers and make their capacity available to many customers. Virtualization helps pool that capacity: a hypervisor divides physical hardware into virtual machines with their own operating systems and allocated resources. Containers provide further isolation within a shared operating system, reducing overhead. These technologies allow customers to share infrastructure while maintaining separate environments. The videos emphasize that physical equipment still supports the cloud; its management and delivery as a service change how we obtain and use computing resources.
- The service models determine how customers and providers divide responsibilities. With *infrastructure as a service* (IaaS), the provider manages physical equipment and virtualization, while customers maintain their virtual machines' operating systems, software, and data. *Platform as a service* (PaaS) also transfers management of the operating system and runtime to the provider, allowing customers to concentrate on applications they develop and their data. *Software as a service* (SaaS) supplies a finished application. Google Docs illustrates this model: users create documents through a browser while Google maintains the application and infrastructure. Moving across these models reduces maintenance duties and changes the degree of direct control customers retain.
- Deployment models describe a different choice. A public cloud serves many customers, a private cloud is dedicated to one organization, and a community cloud supports a group with shared requirements. Hybrid arrangements combine environments, allowing organizations to retain some systems locally while using public services. The appropriate combination depends on application needs, existing systems, and where resources must operate.

## Scaling and performance

- Resource pooling, broad network access, on-demand self-service, rapid elasticity, and measured use work together to make cloud capacity adaptable. Authorized users can provision resources when needed, and organizations can expand or reduce capacity as demand changes. Savill illustrates horizontal scaling with a web service that needs ten instances during busy periods and three during quieter periods. A pizza business could scale for Friday evenings or the Super Bowl, then reduce capacity afterward. Predictable changes can be scheduled, while autoscaling can respond to demand. This flexibility connects expenditure to current needs and helps avoid purchasing equipment primarily to accommodate occasional peaks.
- Cloud bills reflect service-specific measures, such as running time, storage capacity, or database activity. Savings therefore depend on selecting appropriate services and releasing unnecessary resources. Self-service also needs governance: permissions, quotas, and policies can control who creates resources and what they may deploy. These controls help organizations obtain computing capacity quickly while managing costs and enforcing operational requirements.
- Location affects both performance and reliability. Providers organize infrastructure into geographic regions, with availability zones separating facilities and supporting systems within a region. Deploying

across zones can protect against localized failures; additional regions can support recovery from wider disruptions. Applications must be designed to use this redundancy. Locating services near users and keeping frequently communicating components close together also reduces latency, since repeated network delays can accumulate during an application operation.

- Storage choices should match how information is used. Block storage provides addressable blocks, file storage organizes shared files and folders, and object storage suits large collections of unstructured content such as images and videos. Hot and cold options accommodate different access frequencies and retrieval needs, with corresponding performance and cost considerations. Content delivery networks distribute content closer to audiences, extending the same geographic principle to faster delivery and improved availability.

## Security and organizational value

- Cloud security follows a shared division of responsibility. Providers protect their infrastructure, while customers must address the responsibilities associated with their chosen services. An IaaS customer still needs to maintain its operating systems and plan workload backup and recovery. SaaS reduces maintenance work, but organizations continue to manage users, access, and their handling of information. Microsoft's defense-in-depth model combines physical protection, identity controls, network restrictions, application security, and data safeguards. Together, these layers support confidentiality, integrity, and availability.
- Identity is central because one cloud identity can provide access to multiple services through single sign-on. Multifactor authentication, device checks, and risk-based policies help protect that access. A zero-trust approach continually evaluates requests instead of assuming that network access establishes trust. Public-cloud use also permits private access to resources: management operations and access to applications or data can be controlled separately through permissions and secure connections. Cloud flexibility depends on configuring these protections alongside the resources being deployed.
- Microsoft's Chesterfield County case study connects these decisions to practical results. The Department of Social Services struggled with fragmented information, handwritten records, and duplicate entries. Adopting Dynamics 365 centralized document and data management and streamlined its workflows. Microsoft reports a 33% reduction in the time required for common tasks and a lower likelihood of documents being duplicated or lost. The case illustrates how a suitable SaaS application can improve daily work. Across the videos, the cloud's value emerges from matching services to organizational needs while making informed choices about responsibility, capacity, performance, and security.